PREETHAM S G

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Education

Manipal School of Information Sciences, MAHE — Manipal, India
Master of Engineering in Computer Science Engineering (Big Data Analytics) — 2025 – Present
CGPA: 7.84

Sapthagiri College of Engineering — Bengaluru, India
Bachelor of Engineering in Computer Science Engineering — 2020 – 2024
CGPA: 7.11

Technical Skills

Programming Languages: Python, SQL, C
Machine Learning & AI: Scikit-learn, TensorFlow, Transformers, FAISS, Generative AI, LLMs, RAG Pipelines, Prompt Engineering, Deep Learning
Data Engineering: PySpark, Hadoop, Apache Airflow, ETL Pipelines, Data Processing, Workflow Automation, Data Integration, Databricks
Backend Development: APIs, API Integration, FastAPI
Databases: MySQL, PostgreSQL, SQL, NoSQL Basics
Cloud & DevOps: Docker, Kubernetes , CI/CD, GitHub Actions, MLflow
Data Analysis: Pandas, NumPy, Matplotlib, Data Cleaning, Trend Analysis, Data Visualization
Tools: Git, GitHub, Linux, Jupyter Notebook, VS Code, Excel

Projects

LLM Powered Product Search System

- Developed a semantic search and question-answering system using Python, Transformers, and FAISS.
- Designed embedding pipelines and vector indexing for efficient similarity search.
- Implemented Retrieval-Augmented Generation (RAG) workflow for context-aware response generation.
- Processed structured and unstructured datasets to improve retrieval quality.
- Reduced response latency by approximately 30% through optimization techniques.

TrafficPulse: Smart Traffic Data Pipeline

- Built an Apache Airflow-based ETL pipeline for ingesting real-time traffic data into PostgreSQL.
- Automated data ingestion, transformation, and validation workflows to improve reliability.
- Optimized pipeline performance and enhanced data processing efficiency.
- Enabled structured reporting and analytical insights using clean datasets.
- Integrated workflow scheduling and monitoring for scalable data engineering operations.

Wind Power Forecasting using Machine Learning

- Built TensorFlow-based forecasting models using time-series wind energy datasets.
- Applied feature engineering and evaluation methods to improve prediction accuracy.
- Used MLflow for experiment tracking and performance monitoring.
- Performed trend analysis and predictive modeling for energy forecasting.
- Applied time-series forecasting techniques and deep learning models for predictive analytics.

Heart Disease Prediction System

- Developed predictive machine learning models using Logistic Regression and Random Forest.
- Performed preprocessing, feature engineering, and model evaluation using Scikit-learn.
- Generated actionable insights from healthcare-related structured datasets.
- Improved model performance through feature selection and hyperparameter optimization.

Certifications

- Oracle Cloud Infrastructure – Certified Data Science Professional (2025)
- Snowflake – Data Engineering Fundamentals
- AWS Cloud Fundamentals